

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
H H T H H H T H H T
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.
5. Given:
Input (x) = 4
Weight (w) = 2
Actual Output = 10
Prediction:
 $y = w \times x$
 1. Find the predicted output.
 2. Calculate the error.
 3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

I OmSwaroop Pannuru(192425275) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Om Swaroop Pannuru

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. MLE of Probability of Heads

Given outcomes:

[H, H, T, H, H, H, T, H, H, T]

Count the outcomes:

- Number of Heads (= 7)
- Total number of tosses (= 10)

For a coin toss, the probability of Heads is estimated using Maximum Likelihood Estimation as:

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total tosses}}$$
$$\hat{p} = \frac{7}{10} = 0.7$$
$$\boxed{\hat{p} = 0.7}$$

Interpretation: The MLE estimate shows that the coin produced Heads in 70% of the observed tosses. Therefore, the estimated probability of getting Heads is **70%**.

2. MLE of Student Passing Probability

Given:

- Total students (= 50)
- Students passed (= 42)

Each student either passes or fails, so this is a Bernoulli experiment. The MLE for the probability of passing is:

$$\hat{p} = \frac{\text{Number of students who passed}}{\text{Total students}}$$
$$\hat{p} = \frac{42}{50} = 0.84$$
$$\boxed{\hat{p} = 0.84}$$

Interpretation: Based on this sample, the probability that a student passes the exam is estimated as **0.84**, or **84%**. Approximately 84 out of every 100 similar students are expected to pass.

3. MLE of Defective Bulb Probability

Given:

- Total bulbs inspected $(= 100)$
- Defective bulbs $(= 8)$

A bulb can be classified as either defective or non-defective. Therefore, the defective-bulb event can be modeled using a Bernoulli distribution.

$$\hat{p} = \frac{\text{Number of defective bulbs}}{\text{Total bulbs}} \\ \hat{p} = \frac{8}{100} = 0.08 \quad \boxed{\hat{p} = 0.08}$$

Interpretation: The estimated probability that a bulb produced by the factory is defective is **8%**. In practical terms, about 8 bulbs may be defective for every 100 bulbs manufactured.

4. MLE of Rolling a 6

Given:

- Total die rolls $(= 60)$
- Number of times 6 appears $(= 18)$

The MLE estimate is the observed frequency of rolling a 6:

$$\hat{p} = \frac{\text{Number of sixes}}{\text{Total rolls}} \\ \hat{p} = \frac{18}{60} = 0.3 \quad \boxed{\hat{p} = 0.3}$$

Interpretation: The estimated probability of rolling a 6 is **30%**. A fair die has probability $(1/6 \approx 16.67\%)$ of producing a 6. Since 30% is much higher, the die appears to be biased in favour of 6.

5. Prediction, Error, and Weight Update

Given:

$$x=4, \quad w=2, \quad \text{Actual output}=10 \\ \text{Gradient}=4, \quad \eta=0.01$$

where η is the learning rate.

5.1 Predicted Output

Forward propagation calculates the output using:

$$y=w \times x \\ y=2 \times 4=8 \quad \boxed{\text{Predicted output}=8}$$

Interpretation: Using the current weight of 2, the model predicts an output of 8.

5.2 Error

The difference between the actual output and predicted output is:

$$\text{Error} = \text{Actual output} - \text{Predicted output} \\ \text{Error}=10-8=2 \quad \boxed{\text{Error}=2}$$

Interpretation: The model prediction is 2 units lower than the actual output. Therefore, the model needs to learn and adjust its weight.

5.3 Updated Weight Using Gradient Descent

The gradient descent weight-update rule is:

$$w_{\text{new}} = w - \eta \times \text{Gradient}$$

Substitute the values:

$$w_{\text{new}} = 2 - (0.01 \times 4) \\ w_{\text{new}} = 2 - 0.04 \\ w_{\text{new}} = 1.96 \quad \boxed{w_{\text{new}} = 1.96}$$